Design and Web Development of E-Voting System

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***Abstract* – Voting plays a very important role in electing a candidate in the government and private organizations these days. Voting system's integrity and privacy are the main challenges in developing a voting system. The person who is voting a particular party should not be known by everyone which is a concern to privacy of the voter and the voting system should give correct results which shows the integrity of the voting system. One single vote may be responsible to change the entire result of the polling. So, voting system should be designed in such a way that each vote should be given its own importance and that should not get tampered by the hackers. So to solve this problem we are developing a full stack web application on voting system which does not compromises with integrity, privacy and security. This project has been developed using a server-side scripting language PHP. As the name server-side suggests the code which has been written cannot be accessible to the client thus it ensures privacy and security.**

Key Words – Voting, PHP, SQL, Polling, Ballot box voting

I. Introduction

We are developing a full stack project on voting system which ensures integrity and privacy. Voting system's main objective is to give voter a platform to vote to his favourite party and also make participants to contest in the election. Voting system is not required when the polling is happening for a smaller number of voters, they can conduct election by just raising their hands but the problem with this kind of election is everyone will be knowing everyone's vote. This may lead to many problems like, if the person winning the election will be knowing the candidates who supported him and the candidates who did not support him, this leads to separation of the organization into two groups so, this is the reason why privacy became a very important objective in polling.

When it comes to polling for a huge number of voters then at this point, election has one and only option as voting system as it cannot be done with raising hands so, privacy is implicitly handled by using voting system. If we use voting system in both cases then, this solves all the problems. So, even if we consider for a polling which involves a smaller number of voters and the polling which involves a large number of voters in both cases voting system is mandatory.

This voting system is an essential part of election in both government organizations and private organizations. As far as government organizations are concerned, elections are backbone of democracy because people will decide their leader depending on their views, performance of that party in previous rule etc. People of the nations are getting empowered by democracy to choose their own leader, which indirectly makes government to do justice to their people. So, this way voting system indirectly provides justice to the people of the nation.

Polling is also required in private organizations to elect their chairman in companies, student elections in educational institutions to elect their head boy, internal elections in a political party to elect their president, employee elections in their companies to elect present to represent the company's officials on behalf of all employees. In this way not only in government organizations but also in private organizations voting system is necessary.

This voting can be done can also be done using manual polling in which voter will choose their favourite party by

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making a mark on the paper and submit it in the ballot box. But there are some limitations with that kind of polling some of them are tampering of votes can be cone very easily by submitting more than one response, by filling ink into ballot box etc and even while counting the votes as it is done by employees of voting authority human errors are very common which leads to incorrect results in election which are an integrity issues to the election.

When voting becomes online voting system voter can vote only once which overcomes the limitation of ballot box voting system. Tampering is impossible in voting system which is a most common issue with ballot box voting system. In online voting system the counting will the done by the computer in which human errors are impossible to occur. As far as manual voting is concerned declaring results take time but when it comes to online voting system it can be done with just one click, in this way online voting is comparatively a better platform for polling than manual ballot box voting system.

II. Literature survey

From the paper [1] Ramya Govindarai has developed a full stack project on voting system which has been implemented with C# as a programming language and used Microsoft SQL server 2012 as a query language and he also says that his system gives large database support which also checks whether the voter is above 18 or below 18 while doing registration for the polling. They also discussed the negative impacts of electronic voting system and states that online system programmed with C# is the best option. They also says that online voting system is in fact easy to handle compared to ballet box voting system. Form their use case diagram it is clearly understood that voter's email will generate a unique id for every voter which will be used for voting.

From the paper [2] Anisaara Nadaph proposes a online voting system which gives voter two choices for voting they are: one is SMS based voting system and another is Website based voting system. In both the voting systems ultimately, database will get updated. When it comes to SMS based voting system which can be done through SMS with voice authentication which will go to mobile switching center and finally changes the database with the new vote received from the voter. When Website based voting system is concerned the voting is done through finger print which ensures that the person voting is the account holder of that vote and after receiving the finger print from the voter, the number of votes of voter's favorite party will get incremented by one.

From the paper [3] Kashif Mehboob Khan mainly focuses on the blockchain technology in the context of voting system. They have developed a secured online voting system which depends on blockchain technology. They have developed multichain method and in-depth evaluation method in the development of online voting system. That application is developed using Java EE as a programming language and MySQL database which uses SQL query language. He also says that primary objective of his online voting system is performance. The uniqueness of Kashif Mehboob Khan's online voting system is that system can get the data of voter through Microsoft Excel also.

From the paper [4] Voram Bhavan develops a voting system in which consists of two actors as represented in the use case diagram they are, voter and admin. Voter can access the use cases such as login, display candidate, caste vote, voter profile and admin can access the use cases such as login, update candidate, add new candidate, delete candidate, edit voter profile, filter result, generate reports, display candidate, and display voters. The voters will be provided unique voter ids which are combination of numbers and alphabets which makes the id more unique. Voram Bhavan's online voting system also consists of admin portal in which candidate list will be available at the right part of the page and on the left part of the page status of the voter whether he is online or offline and under main navigations candidates , voting , voter are available and settings are available at the bottom of the navigation column which is available on the left side of the page.

From the paper [5] Emad abu-shanab stresses the importance of voting system in the context of democracy.In his study he considered a small organization which is a university and he implemented his online voting system practically there and make that university get familiar with the online voting system by making them understand in how many ways the online voting system is more advantageous than manual ballot box voting system.

From paper [6] Rajesh M. Ghadi uses finger print based voting system and states that finger print based voting system is more reliable compared to other online voting systems and manual ballot box voting system. Rajesh M. Ghandi uses minutiae-based algorithm for finger print detection. This algorithm is sub divided into again two different algorithms they are minutiae extraction and minutiae matching. Minutiae extraction is the process in which finger prints are extracted from the voter and when it comes to minutiae matching which is a process in which the finger prints of the vote will be taken as the input and checks whether the finger prints match with the previously taken input from the voter, if matches that vote will be considered or else the vote will not be considered. He used MYSQL database, NetBeans IDE and for testing XAMPP server.

From paper [7] Anitha A. Lahane uses blockchain technology for online voting system. He uses a number of blockchain frameworks in the development of online voting system. From his use case diagram, it can be clearly understood that end use will have interaction with admin and voter login which is connected with voter login and this voter login is connected with caste vote. And this admin us case is connected with admin panel which is connected with candidate management, voter management, results and view ledgers. This

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online voting system primarily OTP as to check the authentication of the voter in online voting system.

From the paper [8] Anne-Marie Oostveen have done a research study on online voting system and have answered many questions related to media effects, reliability of the system, privacy of the online voting system, learning effects, usability and organisational aspects in online voting system when compared with manual ballot box voting system.

From paper [9] and paper [10] Uzma Jafar says that online voting system is the most updated technology which is having many open research challenges. Uzma Jafer in his electronic voting system uses blockchain technology. As described in his paper the core components of blockchain technology are node, consensus, miners, chain, chain, transaction and also says that blockchain can transform the electronic voting system to next level. And also concludes that blockchain voting system is comparatively working better than traditional voting system.

From paper [11] Goutham R says that finger print based voting system and also uses Aadhar card. Their main focus is on security of the voting system. As soon as the system receives the vote from the voter the vote in the database will get incremented. During the initial stage of voting process, they are using biometrics for authentication and simultaneously verifies the age o the voter implicitly. This online voting system can be use from any point of the world.

From the paper [12] Aakash has developed a full stack project on online voting system. He uses HTML, CSS, Java Script, MySQL in the implementation of online voting system.

From the paper [13] Subhangi D. Dhane says that there are two different kinds of voting systems available in india today they are manual ballot box voting system and another is electronic voting machine which is popularly known as EVM. He proposes an online voting system in which voting can be done in a more secured manner using finger print sensor , face recognition sensor and QR code scanner and concludes that this approach is far more better than previous two approaches.

From the paper [14] A.C. Santha Sheela develops a voting system based on homomorphic encryption technique. The vote will be given through finger print which will later get encrypted to ensure privacy. Later the number of votes to which the voter has been voted will get incremented by one.

From the paper [15] A.Aravindhan primarily focuses on voting with face recognition in which he uses deep learning techniques such as convolutional neural networks by which the face will get recognized by the system and thus voting will be done.

III. PROPOSED METHOD

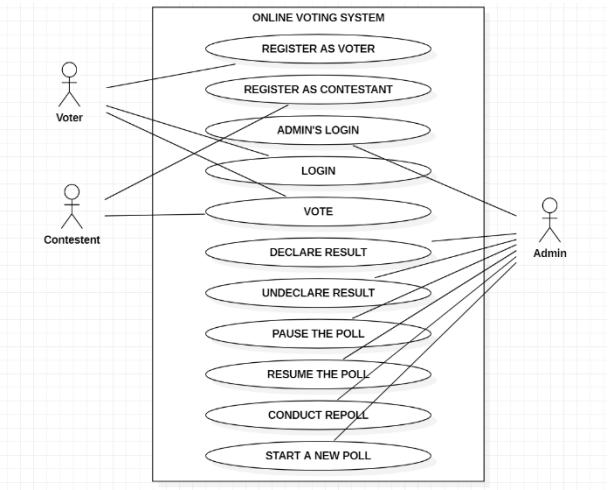

Fig. 1 Use case diagram of online voting system

The above figure 1 shows the use case diagram of online voting system which consists of three actors they are voter, contestant and admin. The use cases are register as voter, register as contestant, admin's login, login, vote, declare result, undeclare result, pause the poll, resume the poll, conduct repoll and start a new poll. Voter is the person who have access to use cases register as voter, login , vote. Contestant will have access to register as contestant, vote. Admin will have access to admin's login, declare result, undeclare result, pause the poll , resume the poll , conduct repoll and start a new poll.

Register as a voter enables voter to register his details such as password, username and his image to the database. Login checks whether the username and the password given by the user matches with the database. If the given information matches correctly with that of the registered information the system allows user to vote. Register as a contestant allows the contestant to register in the election. Admin's login allows admin to login into the admin's panel. Vote allows voter to voter to his favourite party. Declare result allows admin to declare the result, undeclare result enables admin to undeclare the result, pause the poll enables admin to pause the present ongoing poll, resume the poll enables the admin to resume the previously paused poll, conduct repoll enables admin to conduct the poll again, start a new poll enables admin to start a new poll.

This voting is done by using PHP as a server scripting language and for frontend HTML, CSS and bootstrap has been used. PHP is server-side scripting language which especially ensures privacy of the code. The code written by using PHP cannot be viewed by the clients even if they inspect the web page by right click. So, database access will be given to the system with PHP by giving username and password in the code and that data will be stored in the form of variables and used whenever required.

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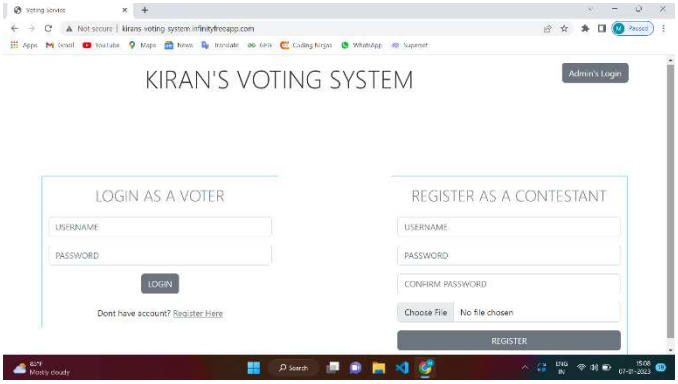

Fig. 2 Home page of online voting system

The above figure 2 is the home page of online voting system which contains login, if the voter did not register, he can click on “register here” which is available below the login button. Home page also contains registration form for contestant which can be used by the contestant to register in the election as a participant in which he need to give his username , password and his image. For voter’s login he needs to give username and password to login. Home page also contains admin’s login which is used by the admin to login to his admin panel to perform admin operations.

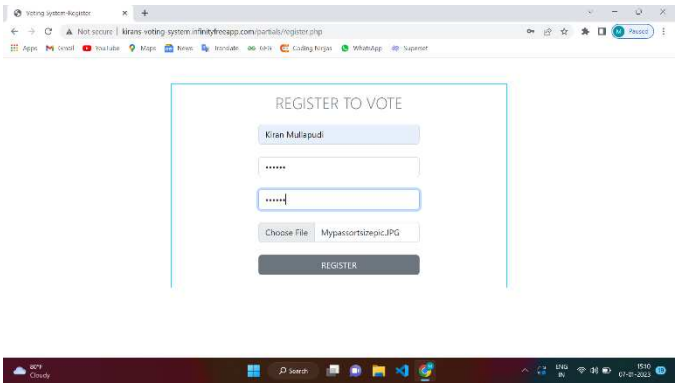

Fig. 3 Registration page for voter

Registration page for vote is given in the figure 3 which is the first step of the voter in the voting process which takes username , password and his image and registers his name for voting.This actually adds a new row to the table in the database which is having user information.In furure this information will be used for verification while loging in for voting.The image which has been uploaded will be stored in a folder in the web server.We will be using SQL as a query language to fetch the data from the database.

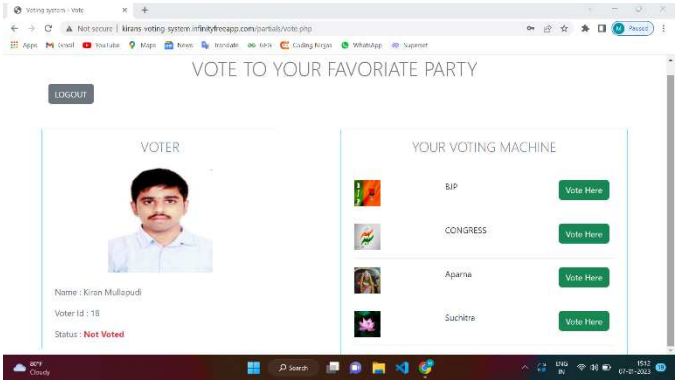

Fig. 4 Voting page before voting

The above figure 4 is the voting which will be displayed after login.Here on the left side you can find the voter’s information such as his image, voter’s name, voter’s id, voter’s status whether his is voted or not.In the figure 4 it is showing not voted in red color because he just logged in.On the right side you can find voting machine which contains the contestants of the election in which voter need to choose his favouriate party. Every party will be given “vote here” button in green color. In the database under status 1 represents voted and 0 represents not voted, as long as 0 is the status in the database “vote here” button will bw there in blue color and the status below the voter’s image will be not voted.On the top left corner logout button is available which allows user to logout from the account and returns to the home page.Here when voter clicks on logout the session which has been created using PHP will get destroyed.Session in PHP stores the information of present voter in session variable.

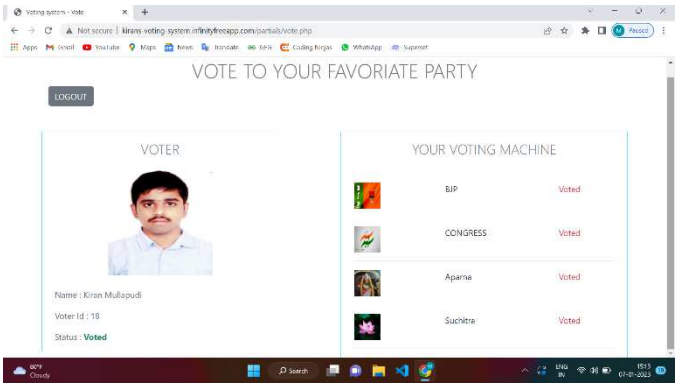

Fig. 5 Voting page after voting

The above figure 5 represents the voting page after voting which means after the the voter have clicked on vote here button.In the figure 5 “vote here” button has been converted into a red color text “voted” , which means the status of the voter has been changed to 1 from 0 in the database.The status in the voter’s details has been converted to voted which previously was not voted.Here we can clear observe that our online voting system is allowing user to vote only once because one the voter has clicked on “vote here” the page will get

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refreshed into the page shown in figure 5.So , in this way our online voting system allows voter to vote only once.

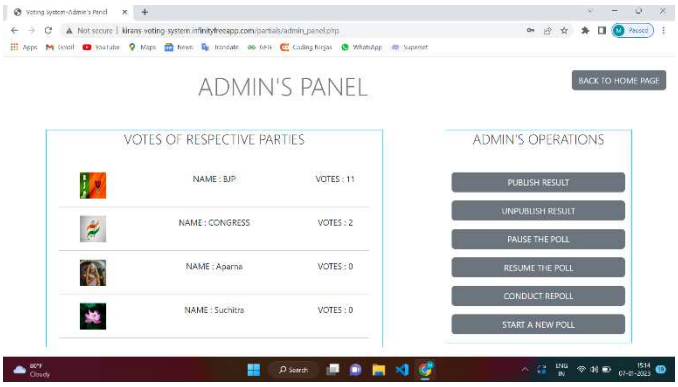

Fig. 6 Admin's panel page

The above figure represents admin's panel which is accessible only to the admin. Admin needs to initially click on the admin's login button available in the home page of the online voting system website then the website redirects to admin's login page where he need to give to login credentials, if the credentials are correct then it will take to the page shown in the figure 6 which is admin's panel. On the left side of admin's panel page votes of respective parties are available and the table is arranged in the descending order of the number of votes gained by the parties. On the right side of the admin's panel admin can find his operations publish result by which he can provide the result on the home page. Unpublish result is the operation by which he can undeclare the result. Pause the poll is the operation by which admin can pause the poll which means if any voter tries to login, it will not allow voter to login by showing a alert message. Resume the poll is the operation by which admin can resume the poll. Conduct repoll is the operation by which the status of every voter will become not voted and the votes of all parties will become zero. Start a new poll is the operation by which all the parties will get deleted and all the voter's status will become not voted.

IV. Conclusion

The human errors which are usually caused by manual counting in ballot box polling are successfully overcomes here in online voting system. Online voting system has less chances of getting tampered compared to the ballot box voting system. As discussed earlier in the proposed method admin will be given complete control to manage voting process which involves several admin operations. These operations can be performed by a single click which is an advancement to ballot box voting system where election organization need to spend a long time to declare result as every thing depends on their employees. In this way online voting system overcomes the limitations of ballot box voting system. From this project we would like to conclude that PHP is the best server-side scripting language that can be used which privacy is the main challenge and MYSQL database also a best choice for this project. The admin has been given complete control over voting platform which has been done with PHP.

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